TRANSITION FROM ACADEMIC TO BUSINESS LIFE

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ABOUT MY-SELF & CHANGE FROM ACADEMY TO BUSINESS

ABOUT MYSELF

Full Name: **Vu** Truong Thanh

DoB: 1976 November 09th (50 yrs old)

Education

1994~1999: Undergraduate from Hanoi University of Science and Technology (Vietnam)
Major: Networking and Telecommunications
2002-2011: Master and PhD from Waseda University (Japan) in IT and Networking
Major: Networking and Telecommunications

Work experience

- Researcher (PTIT-Vietnam, 1999-2005)
- Research Associate (Waseda University, 2007-2010)
- Lecturer on Communications Topics (Post and Telecommunications Institute of Technology -Vietnam, 2012-2015)
- System Integrator/ICT management consultant (2012-2015)
- **Senior Manager @ Hanoi Rep. office (Sept 2015~Now: 10 yrs)**

MY TRANSITION FROM ACADEMIC TO BUSINESS WORLD

Teaching/Researching
~2015

Communications
topics

Consultant
(2012~2015)

System integration
for large ICT systems

Business
(2015~)

- Development of
Infrastructure Business
for a Japanese
Group

MY TRANSITION FROM ACADEMIC TO BUSINESS WORLD

Why:

- Fun fact: Teacher salary in Vietnam is rather low (~400USD/month @ 2015)
- Can complement income by doing project research or consultant
  - I prefer solving real world issues so I choose to be consultant -> too time consuming

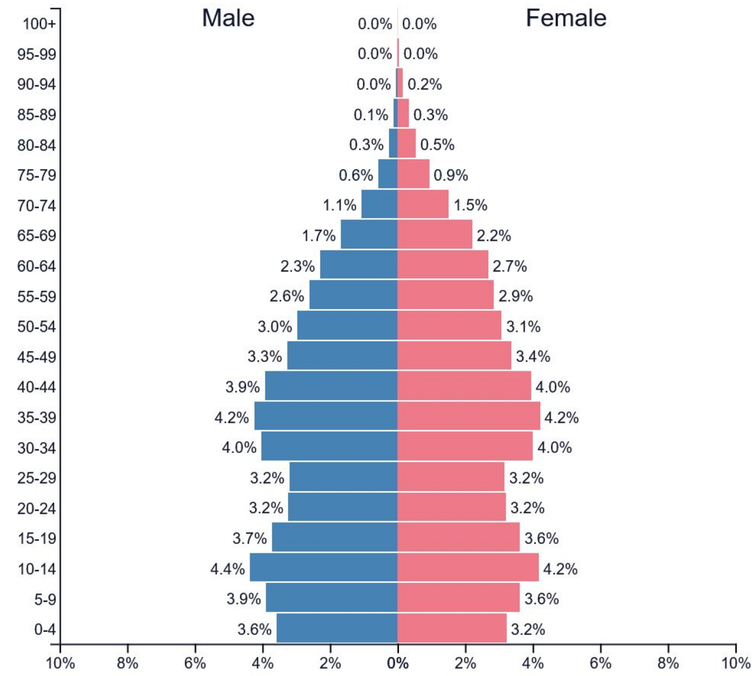
Advantage for someone with academy background

- Having systematic research approach from academy, are **more time consuming** when solving problems in real business world, but help to provide **a more thorough solutions** with credible test cases

If I return to the teaching career in the future, I can give better carrier advices to students

ABOUT VIETNAMESE EDUCATION SYSTEM & CURRENT HOT TOPICS

VIETNAM AT A GLANCE

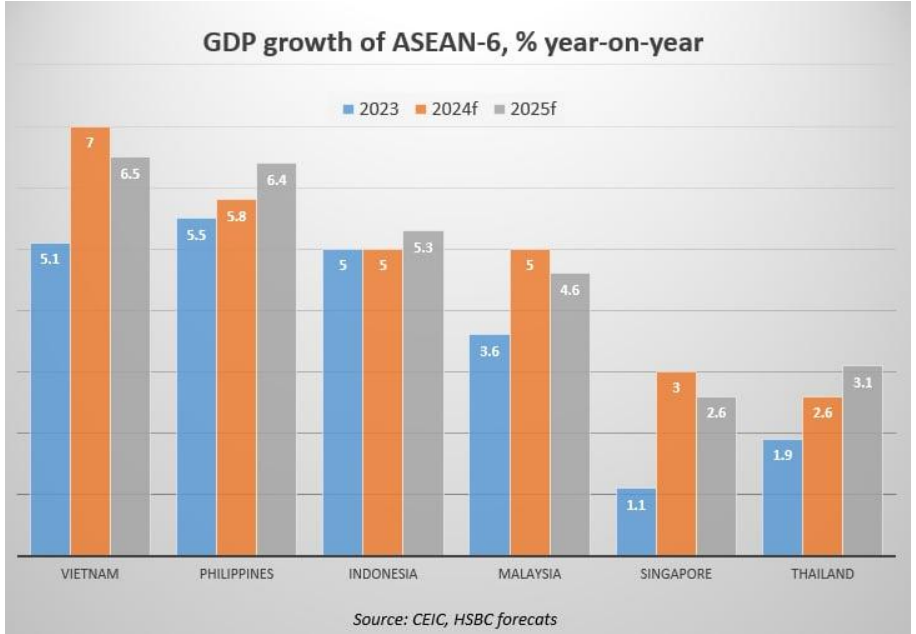


Population: 101 mil., golden pyramid

Area: 331,000 km²

GDP/person: 4800USD

Fast growing economy based on FDI (low skill cheap labor) → moving toward hightech and science



VIETNAM EDUCATION SYSTEM

12 years of general education

- 5 years of elementary (compulsory)
- 4 years of secondary (compulsory)
- 3 years of high school

Undergraduate

- 4 years for bachelor
- 4.5~5 years for engineering
- 6 years for medicine

Higher education are similar to other countries

MAIN CHARACTERISTICS OF VIETNAMESE EDUCATION SYSTEM

Parents have high expectation of kids, would push kids hard to excel in school

Heavy on knowledge, less on critical thinking

- Some remnants from the Soviet system mixing with result-based East Asia style
- Lots of homework, tutoring school
  - pupils have to do model problems over and over again in order to memorize how to solve a specific problem with will come up in the final examinations

Undergraduate courses are reducing textbook teaching, adding more project-research time

- Before: re-training after enter the workforce
- Now: more readiness to perform on job

HOT EDUCATION AND RESEARCH TOPICS

Five major areas

- Semiconductors
- artificial intelligences
- mart factories
- smart cities
- digital technologies

HR Development Target:

- Training 50,000 semiconductor professionals
- 5,000 AI specialists by 2030
- Establishing key research laboratories

NVIDIA

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Press Release

NVIDIA to Open Vietnam R&D Center to Bolster AI Development

December 5, 2024

Home › Regional › Asia-Pacific › Southeast Asia's Latest Talent Visa: Vietnam Opens Its Doors

Southeast Asia's Latest Talent Visa: Vietnam Opens Its Doors

By **Ray de Bono** 27 November 2025


9 VIETNAMESE UNIVERSITIES
MAKE 2025 GLOBAL RANKINGS
(Times Higher Education released its rankings on Oct. 9, 2024)
RANKINGS
501-600
601-800
801-1000
1201-1500
1501+
1 The University of Economics Ho Chi Minh City
2 Duy Tan University
3 Ton Duc Thang University
4 Hanoi Medical University
5 Ho Chi Minh City Open University
6 Vietnam National University, Hanoi
7 Vietnam National University, HCM City
8 Hanoi University of Science and Technology
9 Hue University
Source: Times Higher Education (THE)
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ABOUT MY CURRENT ENGINEERING INTEREST

BATTERY ENERGY STORAGE SYSTEM USING RECYCLED BATTERIES

- Revive/rejuvenate lead-based and Lithium based batteries
  - Multi loop Multi-stage charging
- Smart battery management unit
  - Combined with Revive/rejuvenate function

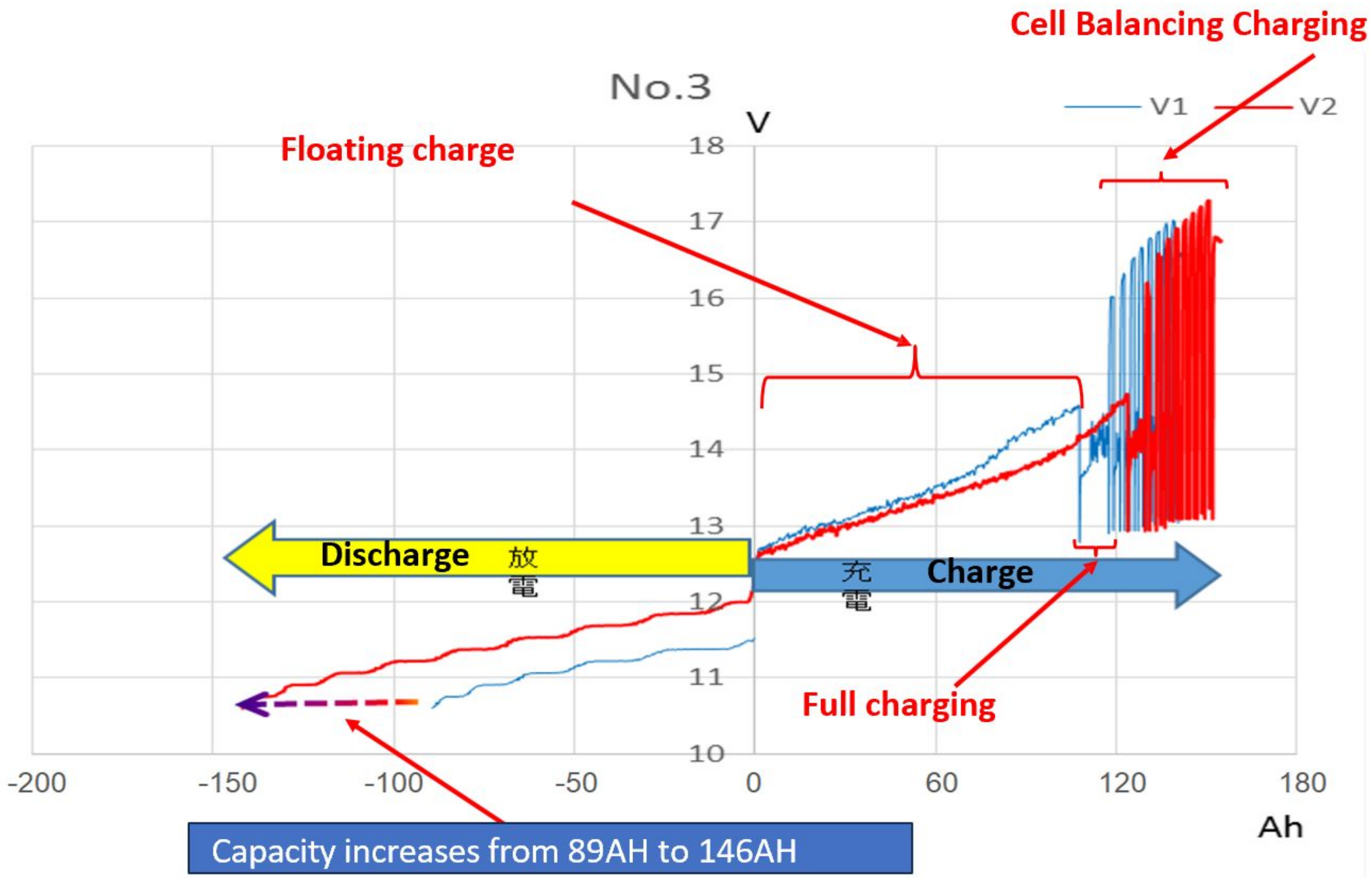
Issues with LEAD ACID Batteries

- SULFATION
- CELL UNBALANCE
- AGING OF ELECTRODE

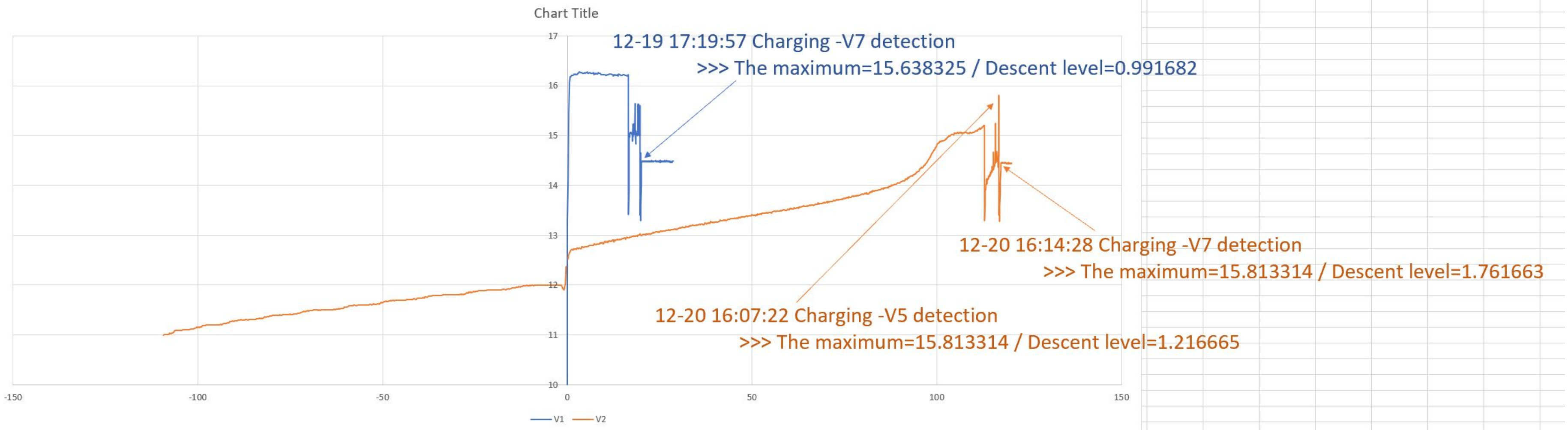
Solutions

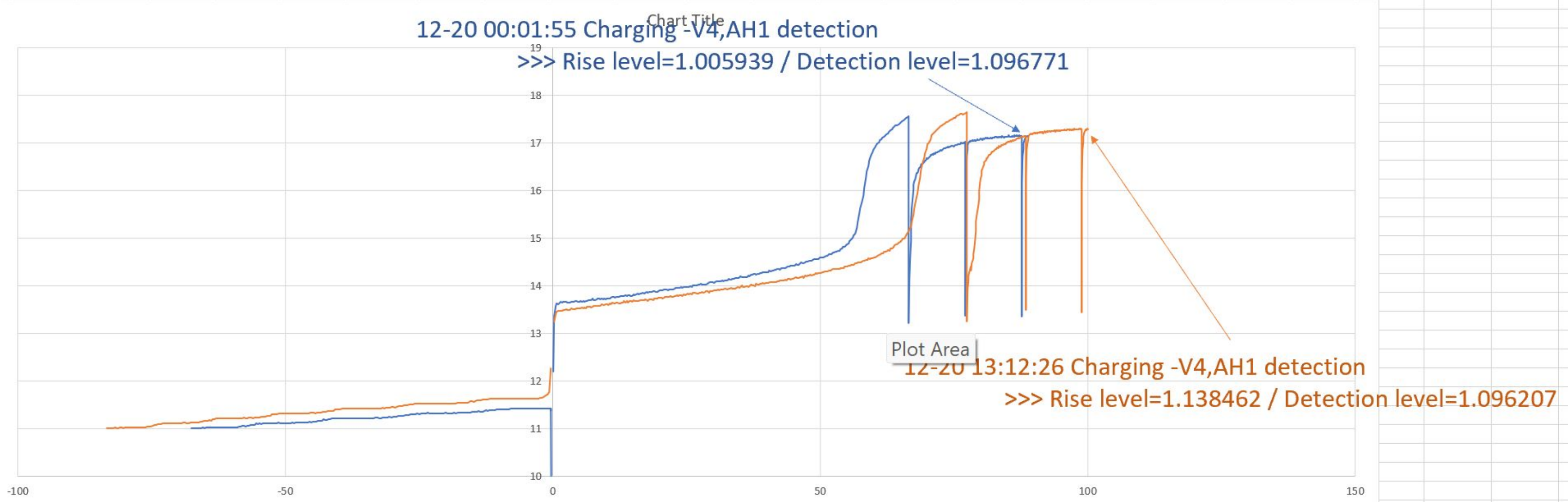
- Using chemical to remove sulfation ⇒ cannot solve Cell unbalance
- Using high power pulse to remove sulfation ⇒ unreliable, cannot solve Cell unbalance

MULTI LOOP MULTI-STAGE CHARGING

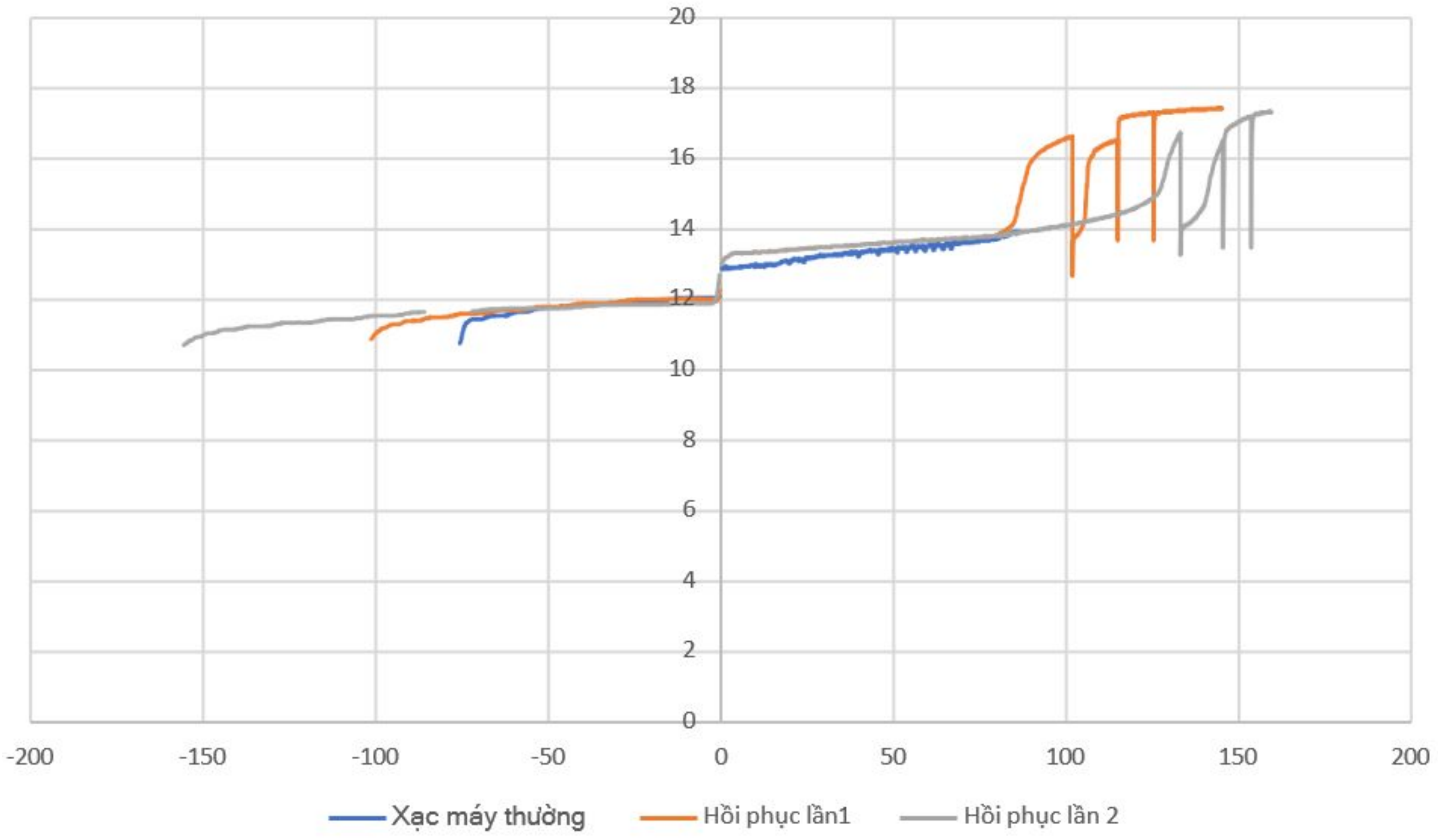


WHY MORE EFFECTIVE: DETECTING ABNORMALITY

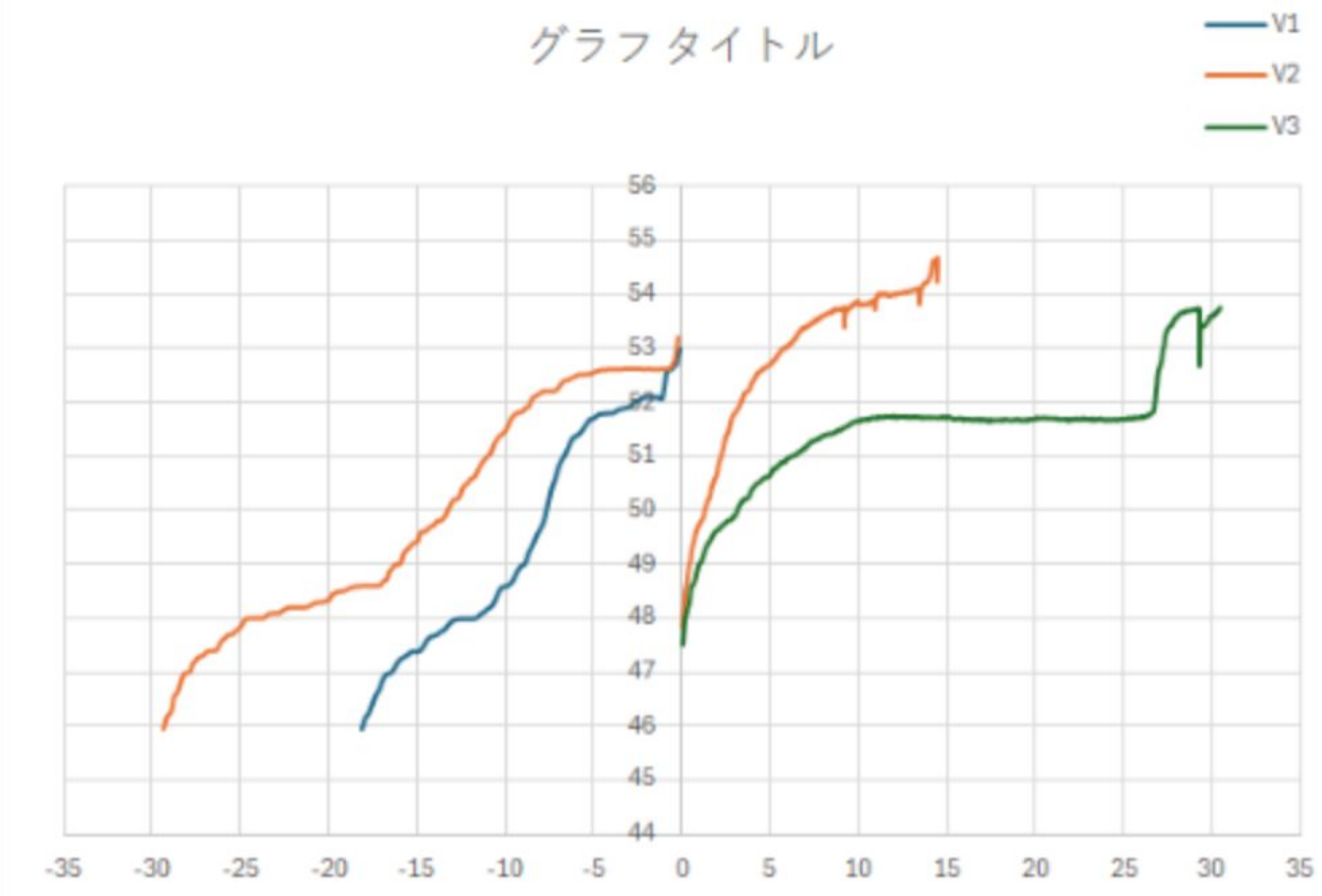




TYPICAL RESULT



TRIAL WITH LITHIUM BATTERY



MY FAVORITE RESEARCH AND EDUCATIONAL APPROACH: MULTI-DISCIPLINE

MY FAVORITE EDUCATIONAL APPROACH: MULTI DISCIPLINE

Multi discipline approach

- Combine knowledges/skills on several areas to invent/develop a useful solution
- Anything you know will become useful at some stage
- Will be more suitable if you are planning to work on real world problems/industry, but can be helpful in theoretical academic research as well
- Example: development of blue-light LED

THE CASE FOR MULTI DISCIPLINE: BLUE LIGHT LED

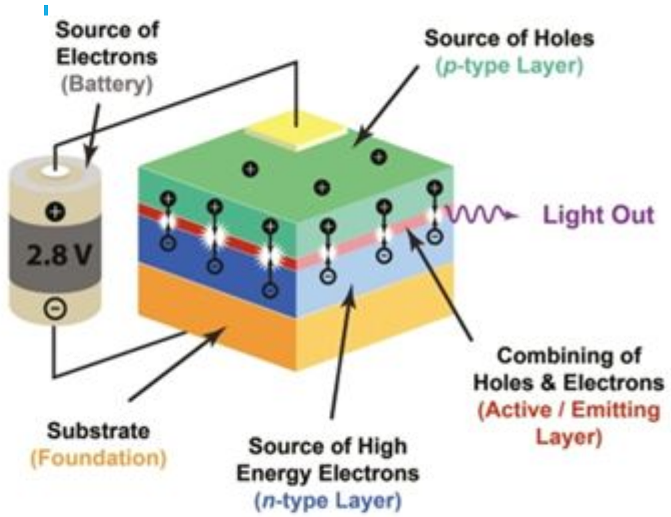
1962: First visible light LED

1968: the red light LEDs became practical and commercially viable

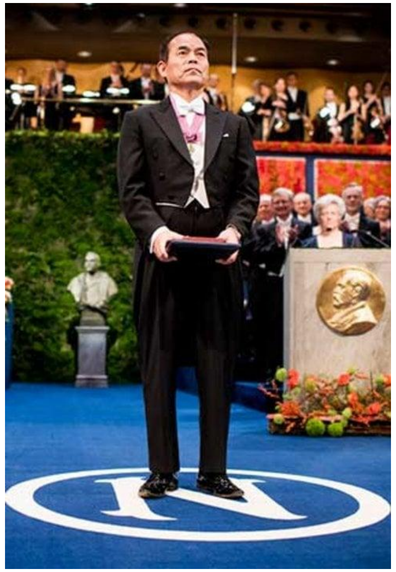
1970s: commercially viable green LED was created

1972: blue light LED initially discovered (very low brightness)

1993: **Shuji Nakamura** from Japan produce blue light LED that is bright enough for lighting purpose



- Research on blue light first as an engineer, then to get his PhD
- Work with Metal Organic Chemical Vapor Deposition (MOCVD) reactor to create mass, clean crystal in U.S in 12 months, enough to tweak the machine to his needs
- At the time, there are two crystal candidates for making Zinc Selenide and Gallium Nitride, Mr. Nakamura chose Gallium Nitride because it was the less preferred medium, which gave more chances for new results (for his PhD)
- Due to **his fluent uses of MOCVD reactor, he can tweak the reactor** to create the necessary crystal to make the blue light LED



He won the 2014 Nobel Prize for Blue Light Invention

AI: CHALLENGE AND OPPORTUNITY FOR MULTI DISCIPLINE APPROACH

Challenge: competing with machine is not easy

Opportunity: it used to cause a lot of time to acquire a sufficient level for a skill, especially in ICT field ⇒ Now you can learn the basics and learn how to use AI to do it professionally

THANK YOU

Q&A